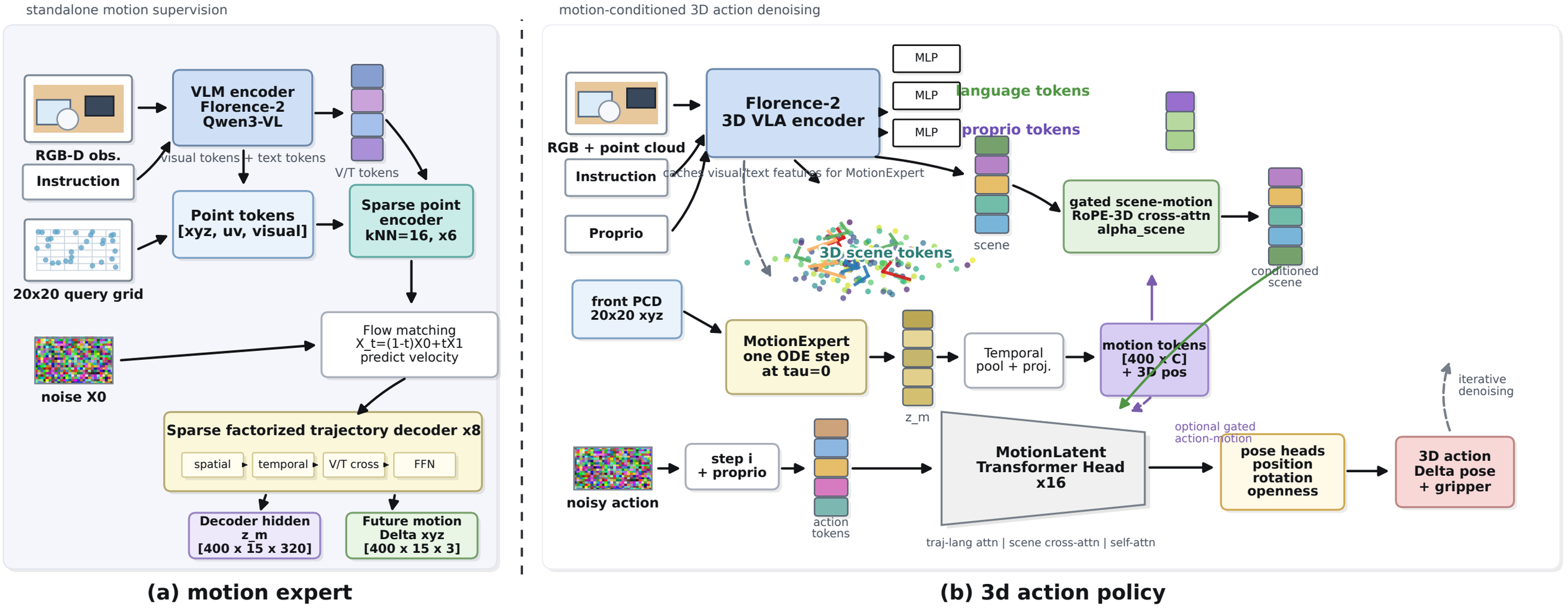

standalone motion supervision
motion-conditioned 3D action denoising
RGB-D obs.
Instruction
VLM encoder
Florence-2
Qwen3-VL
visual tokens + text tokens
V/T tokens
20x20 query grid
Point tokens
[xyz, uv, visual]
Sparse point
encoder
kNN=16, x6
noise X0
Flow matching
X_t=(1-t)X0+tX1
predict velocity
Sparse factorized trajectory decoder x8
spatial
temporal
V/T cross
FFN
Decoder hidden
z_m
[400 x 15 x 320]
Future motion
Delta xyz
[400 x 15 x 3]
RGB + point cloud
Instruction
Proprio
Florence-2
3D VLA encoder
MLP
MLP
MLP
language tokens
proprio tokens
caches visual/text features for MotionExpert
3D scene tokens
scene
gated scene-motion
RoPE-3D cross-attn
alpha_scene
conditioned
scene
front PCD
20x20 xyz
MotionExpert
one ODE step
at tau=0
z_m
Temporal
pool + proj.
motion tokens
[400 x C]
+ 3D pos
iterative
denoising
noisy action
step i
+ proprio
action
tokens
MotionLatent
Transformer Head
x16
optional gated
action-motion
pose heads
position
rotation
openness
3D action
Delta pose
+ gripper
traj-lang attn | scene cross-attn | self-attn
(a) motion expert
(b) 3d action policy